

Contact Management System

A console-based Contact Management System developed in Java using Object-Oriented Programming (OOP) and Data Structures & Algorithms (DSA) concepts.

Features

- Add new contacts with auto-generated unique Contact IDs
- Display all saved contacts
- Search contacts by name or exact phone number
- Update existing contact information
- Delete contacts using Contact ID
- Sort contacts alphabetically by name
- Validate user input using regex and length checks
- Save and load contacts using a local contacts.txt file

Technologies Used

Java • Object-Oriented Programming (OOP) • ArrayList • Java Collections Framework • Collections.sort() • BufferedReader • BufferedWriter • Scanner • JDK

DSA Concepts

The project demonstrates dynamic data storage using ArrayList, searching and filtering, alphabetical sorting using Collections.sort(), string matching, input validation, and file-based data persistence.

Project Structure

```
Contact-Management-System/  
■  
■■■ Contact.java  
■■■ ContactManager.java  
■■■ Main.java  
■■■ contacts.txt  
■■■ README.md  
■■■ .gitignore
```

How to Run

1. Clone the repository:

```
git clone https://github.com/YOUR-USERNAME/contact-management-system-java.git
```

2. Navigate to the project:

```
cd contact-management-system-java
```

3. Compile:

```
javac *.java
```

4. Run:

java Main

Application Menu

```
=====
CONTACT MANAGEMENT SYSTEM
=====

1. Add Contact
2. Display All Contacts
3. Search Contact
4. Update Contact
5. Delete Contact
6. Sort Contacts Alphabetically
7. Save Contacts to File
8. Load Contacts from File
9. Save and Exit
```

Data Persistence

Contact records are stored locally in contacts.txt. The application uses BufferedReader and BufferedWriter to save and restore contact records across sessions.

Project Objective

The objective is to provide a structured solution for managing contact information while demonstrating practical implementation of Java, OOP, DSA, searching, sorting, input validation, and file handling.

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